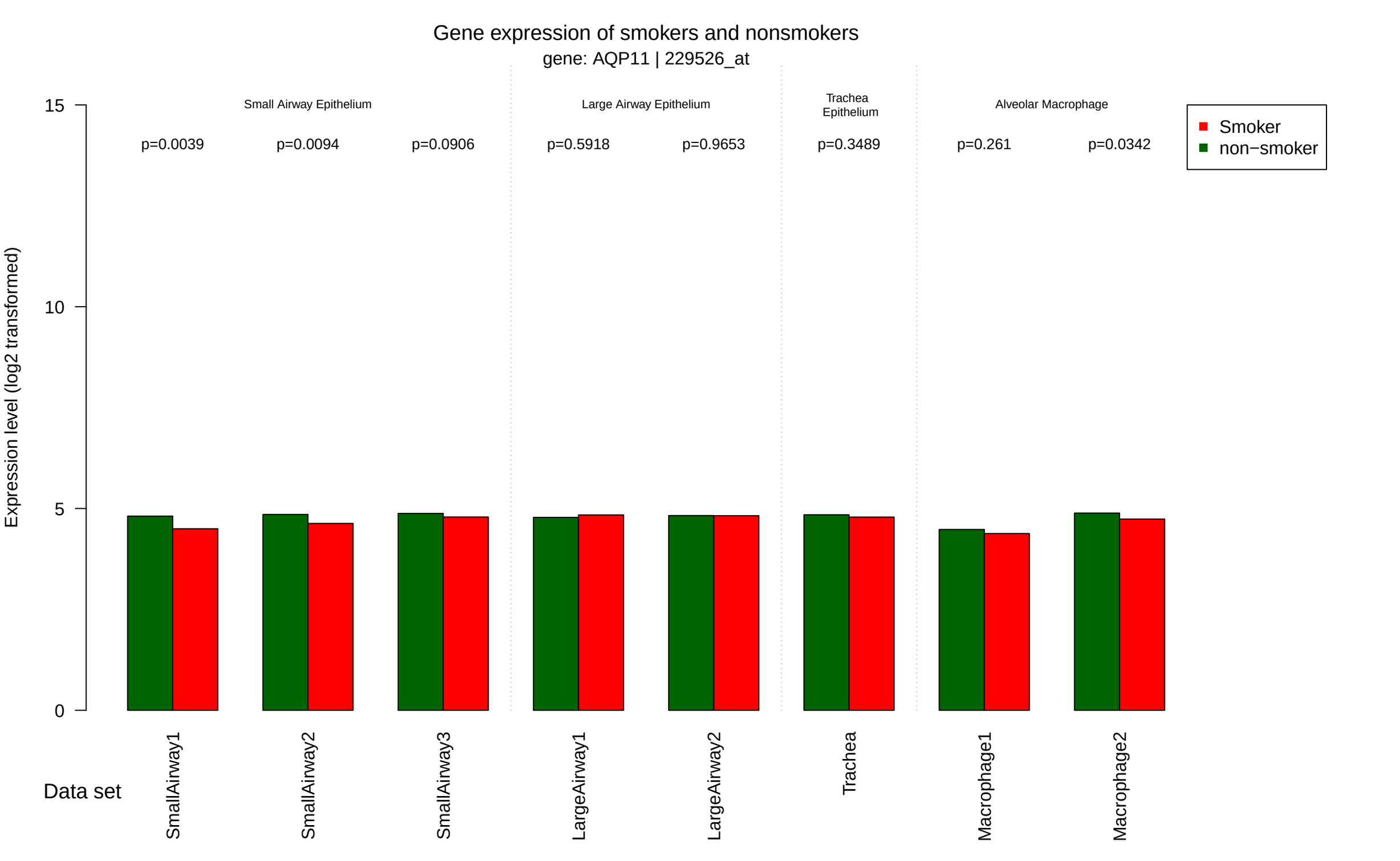
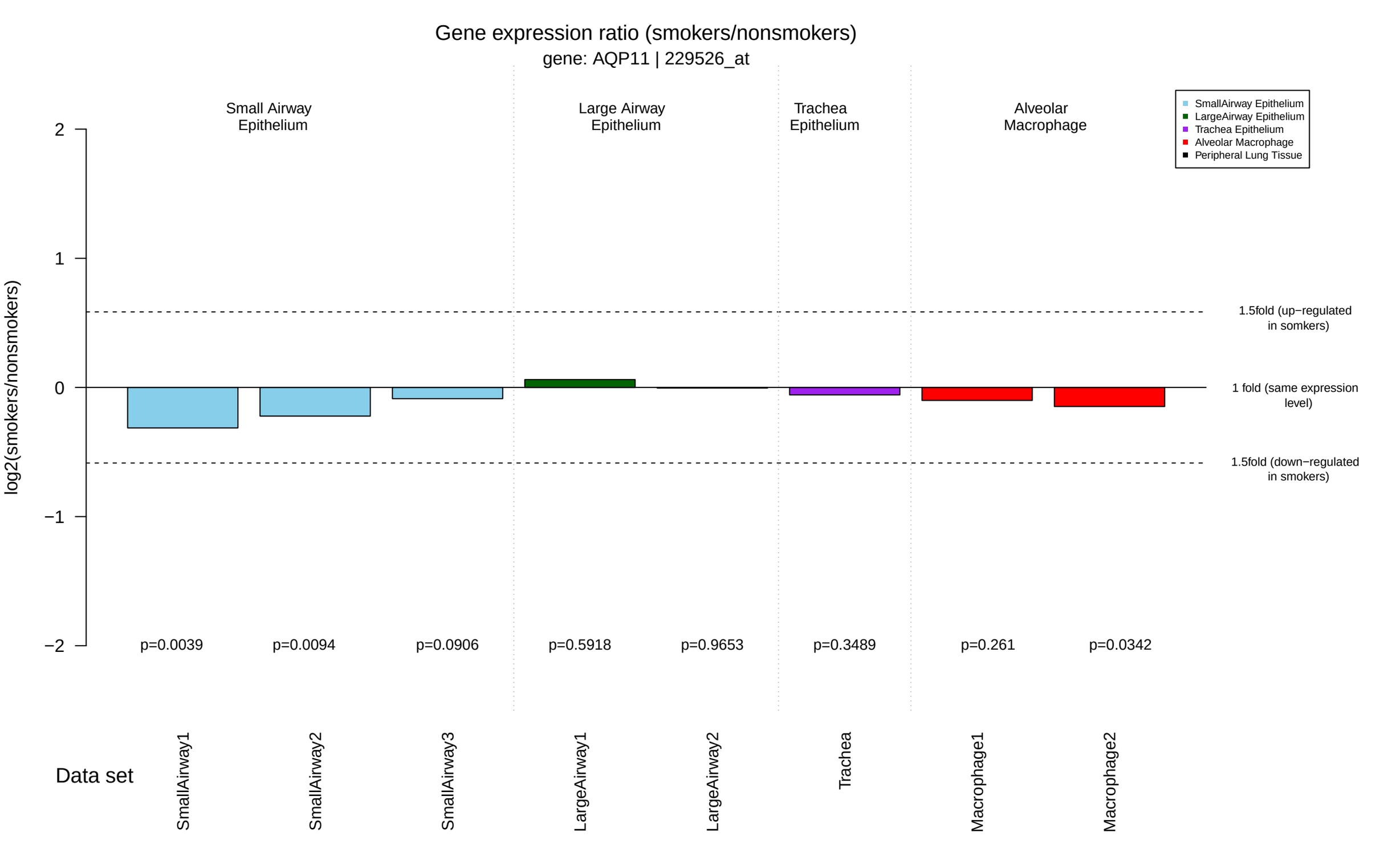
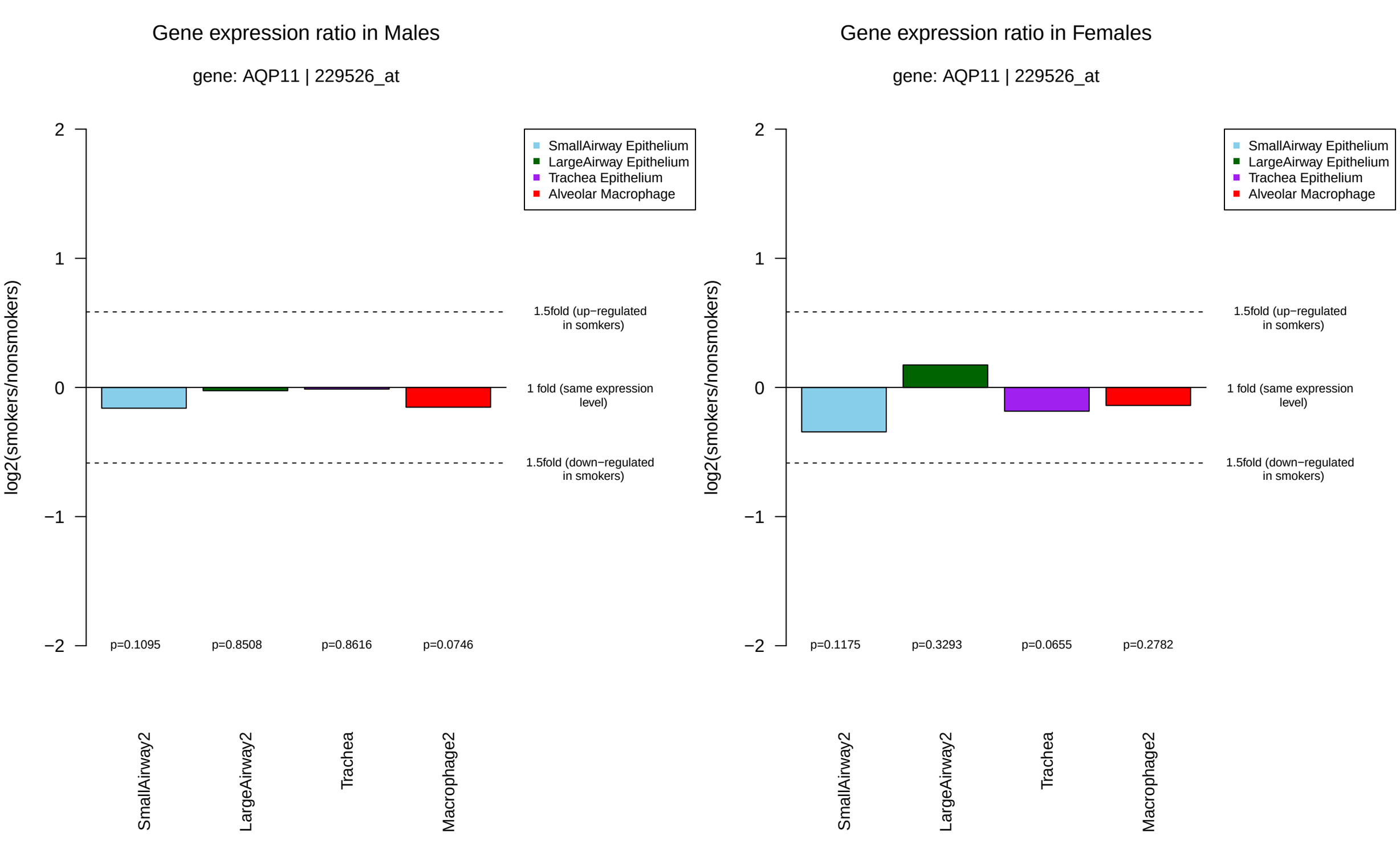
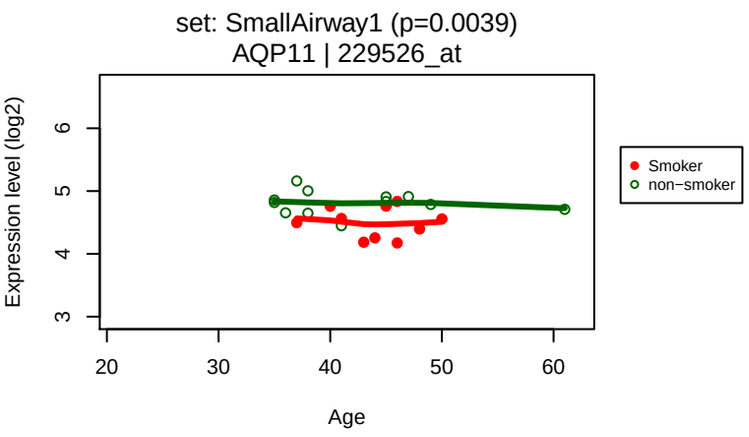
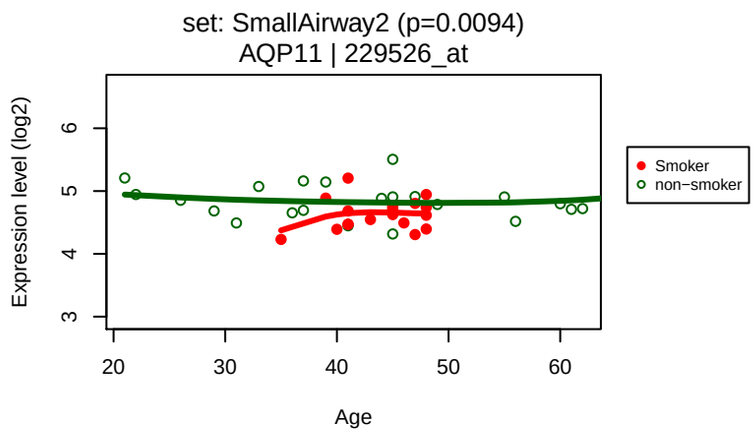
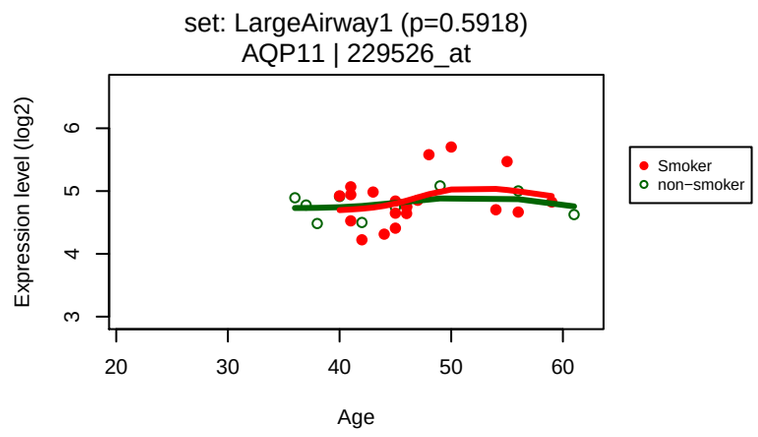

Gene expression of smokers and nonsmokers
gene: AQP11 | 229526_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0039
p=0.0094
p=0.0906
p=0.5918
p=0.9653
p=0.3489
p=0.261
p=0.0342
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

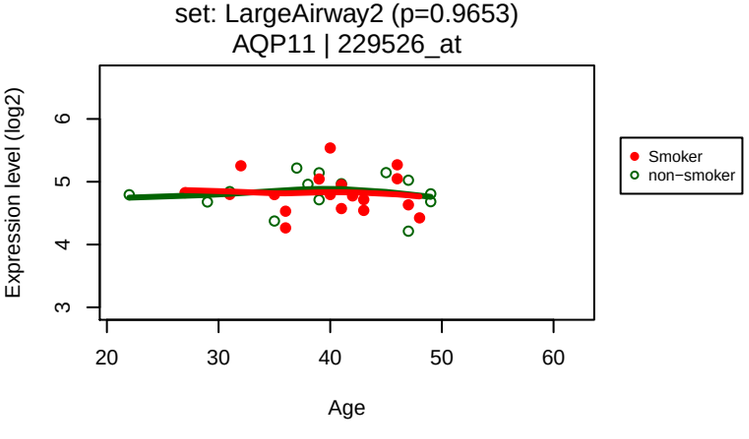

Gene expression ratio (smokers/nonsmokers)
gene: AQP11 | 229526_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0039
p=0.0094
p=0.0906
p=0.5918
p=0.9653
p=0.3489
p=0.261
p=0.0342
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

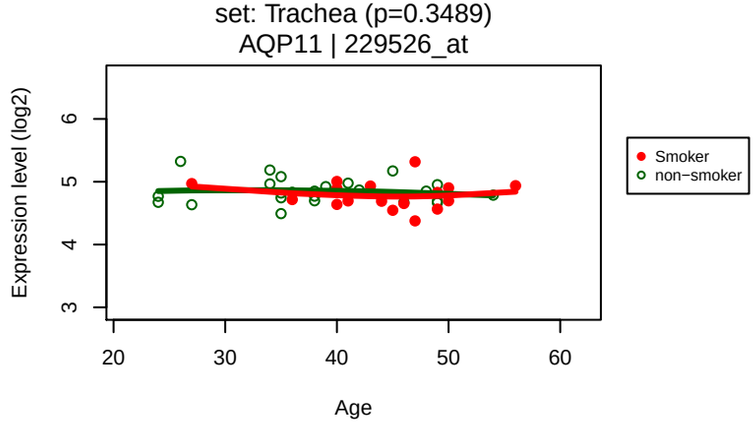

Gene expression ratio in Males
gene: AQP11 | 229526_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1095
p=0.8508
p=0.8616
p=0.0746
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: AQP11 | 229526_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1175
p=0.3293
p=0.0655
p=0.2782
SmallAirway2
LargeAirway2
Trachea
Macrophage2

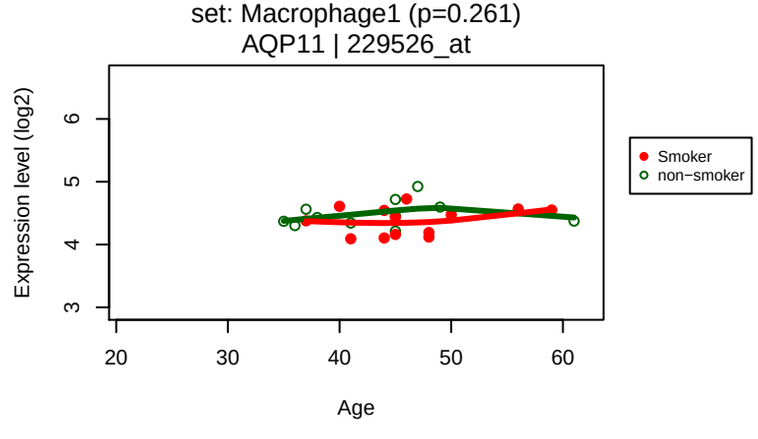

set: SmallAirway1 (p=0.0039)
AQP11 | 229526_at
Expression level (log2)
Age
Smoker
non−smoker

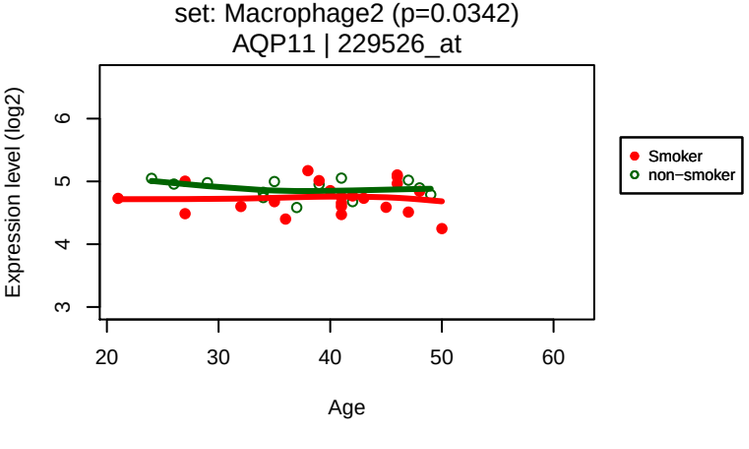

set: SmallAirway2 (p=0.0094)
AQP11 | 229526_at
Expression level (log2)
Age
Smoker
non−smoker

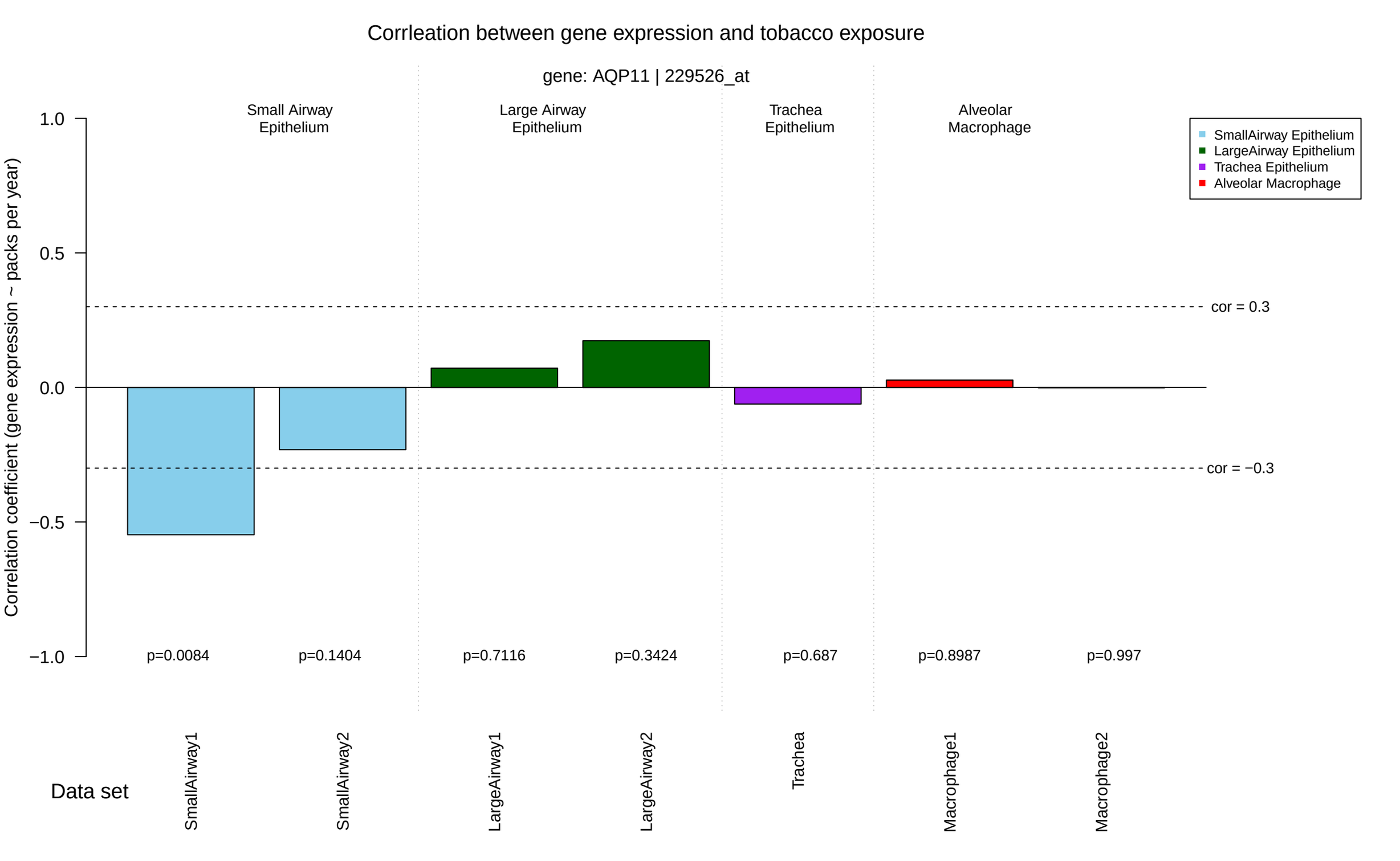

set: LargeAirway1 (p=0.5918)
AQP11 | 229526_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9653)
AQP11 | 229526_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3489)
AQP11 | 229526_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.261)
AQP11 | 229526_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0342)
AQP11 | 229526_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: AQP11 | 229526_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0084
p=0.1404
p=0.7116
p=0.3424
p=0.687
p=0.8987
p=0.997
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2